Paper Title:

Newton’s forward interpolation: representation of numerical data by a polynomial curve

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What it is:

In the mathematical field of numerical analysis, interpolation is a type of estimation, a method of constructing (finding) new data points based on the range of a discrete set of known data points.

Newton's forward difference formula is a finite difference identity giving an interpolated value between tabulated points in terms of the first value and the powers of the forward difference.

Why needed:

Newton's forward difference interpolation can be used to derive numerical schemes for solving initial or boundary-value problems & to find approximate values of the function at intermediate points. It can also be used to predict unknown values for any geographic point data, such as elevation, rainfall, chemical concentrations, noise levels, and so on. Overall it can help in many important calculations of daily life.

How it works:

Use the Newton-Gregory forward interpolation formula to estimate f(2.5) from the following
data:

x	2	3	4	5	6
f(x)	1	2	6	24	120

Solution: Form the forward difference table:

x	y=f(x)	Δy	$\Delta_2 y$	$\Delta_3 y$	$\Delta_4 y$
2	1				
		1			
3	2		3		
		4		11	
4	6		14		53
		18		64	
5	24		78		
				96	
6	120				

With x = 2.5, xo= 2, h = 1 then k = 0.5 and we obtain
f(2.5) ≅ 1 + 0.5 - 0.375 + 0.6875 - 2.0703 = -0.2578.

Limitations:

Although this method/formula is useful but it also has some limitations, or we can say it works under fixed conditions. Such as data should be limited, difference value between the data's should be equal etc etc.

Future Work: